COUNTING HEXAGONAL AND TRIANGULAR POLYOMINOES

W. F. Lunnon†

Atlas Computer Laboratory
Chilton, Didcot
Berkshire, England

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1. Introduction

We have discussed the problem of counting p-minos (polyominoes, animals) on the square tessellation [1]. Here we extend those methods to the hexagonal and triangular tessellations. Our account will be self-contained, but less discursive than before.

A *hex*(agonal)/*tri*(angular) *p-mino* is an edge-connected configuration of p

† Present address: Department of Computing in Mathematics, University College, Cardiff, Wales.

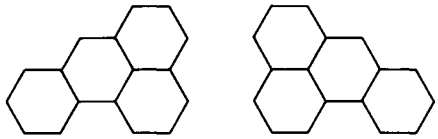
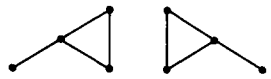


Fig. 1



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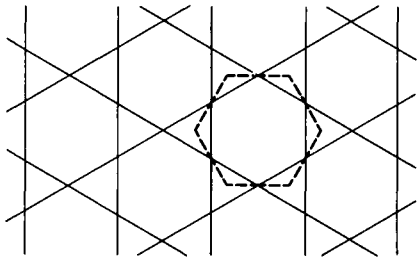

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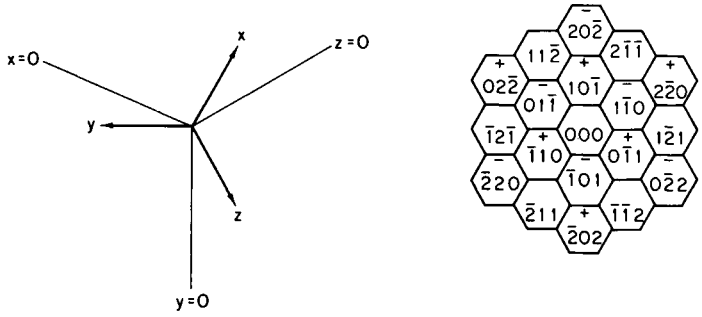

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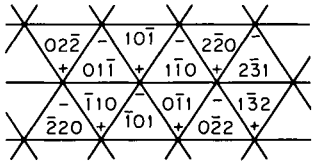


Fig. 5

cells from the appropriate plane tessellation. A *fixed p-mino* is an equivalence class under translation; a *free p-mino* is a class under all symmetries of the lattice. Figure 1 shows a pair of distinct fixed 4-minos belonging to the same free 4-mino, abbreviated in Fig. 2. We attempt to evaluate the totals $HX(p)$, $HE(p)$, $TX(p)$, $TE(p)$ of fixed/free hex/tri p-minos: our results are in Section 5.

Working with these tessellations is facilitated by a sanitary system of coordinates, which we now develop.

Consider the solid tessellation of cubes with a cube center at each integer point $\mathbf{x} = (x, y, z)$, and a plane $x+y+z = 0$ cutting it. This plane intersects each $x+y+z = 0$ cube in a hexagon, and each $x+y+z = \pm 1$ cube in a triangle: the whole tessellation meets the plane in the plane tessellation $6^2 3^2$, of which Fig. 3 is a fragment.

Let us obliterate the triangles and grow the hexagons into the vacancies as shown. The result is the hex tessellation (see Fig. 4). These hexagons correspond one-to-one with the integer triples (x, y, z) such that $x+y+z = 0$.

Now Fig. 4 may be colored in a natural way with 3 colors, indicated by $-$, $\ $, $+$, according as $(x-y)(y-z)(z-x) \equiv x-y \equiv -1, 0, +1$ modulo 3. Let us obliterate the color 0 cells and grow the remainder into the vacancies as before. The result is the tri tessellation (see Fig. 5). A cell is colored $-$ or $+$ according to whether it points up or down. Hence tri p-minos are a subset of hex p-minos: ——

STATEMENT 1. If $p > 1$, the tri p-minos correspond to the hex p-minos on just two colors.

The six neighbors of a hex cell (x, y, z) are

$$(1) \qquad (x, y+1, z-1), \quad (x-1, y, z+1), \quad (x+1, y-1, z),$$

$$(2) \qquad (x, y-1, z+1), \quad (x+1, y, z-1), \quad (x-1, y+1, z).$$

For a color $+$ triangle, (1) are neighbors; for a color $-$ triangle, (2) are neighbors.

2. Bounding Hexagons

The *bounding hexagon* (bh) of a p-mino, hex or tri, is defined by the lines

$$x = a', \quad x = a, \quad y = b', \quad y = b, \quad z = c', \quad z = c,$$

where a' is the minimum of x over all its cells, a is the maximum, and so forth for y, b and z, c. Figure 6 shows a bh, together with the vertex coordinates and the side lengths s_i: since $x+y+z = 0$, we use * to mean "minus the sum of the other two."

For describing the shape of a bh we use these four independent intrinsic parameters: ——the *diameters*

$$A = a - a', \quad B = b - b', \quad C = c - c',$$

and the *skew*

$$K = a + b + c + a' + b' + c'$$
$$= s_1 - s_4 = \text{difference between any pair of opposite sides.}$$

The sides can be expressed in terms of these four as

$$s_1 = a' + b + c = \tfrac{1}{2}(K - A + B + C), \tag{3}$$
$$s_4 = a + b' + c' = \tfrac{1}{2}(-K - A + B + C), \qquad \text{etc.,}$$
$$s_1 + s_5 + s_3 = \tfrac{1}{2}(K + A + B + C), \qquad \text{etc.} \tag{4}$$

We now ask what values of A, B, C, and K are possible. Reflecting and rotating the bh (see Section 3) corresponds to permuting A, B and C and negating K. So we can always arrange, for example, that

$$0 \leqslant K \quad \text{and} \quad C \leqslant B \leqslant A. \tag{5}$$

Furthermore, $|K| \leqslant \min(C, B, A)$: for K is largest compared to C, let us say, in a triangular bh like Fig. 7, and increasing K by 1 inevitably increases C by 1 as well.

Again, since s_1, etc. are nonnegative, by (3)

$$A + K \leqslant B + C, \text{ etc.;}$$

and $s_1 + s_5 + s_3$ is an integer, so by (4) $A + B + C + K$ is even.

To sum up, for a bh of fixed orientation, it is necessary that

$$0 \leqslant K \leqslant C \leqslant B \leqslant A \leqslant B + C - K \tag{6}$$

with $A + B + C + K$ even. We claim that these conditions are also sufficient.

We now ask what further restrictions a fixed value of p entails. Tri p-minos can occupy only 2 colors, which makes the analysis difficult; so we restrict

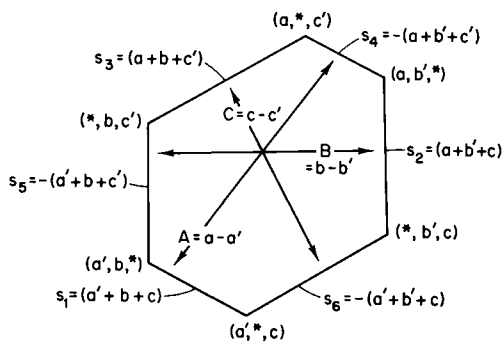


Fig. 6

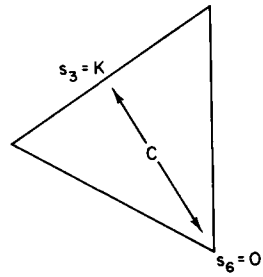


Fig. 7

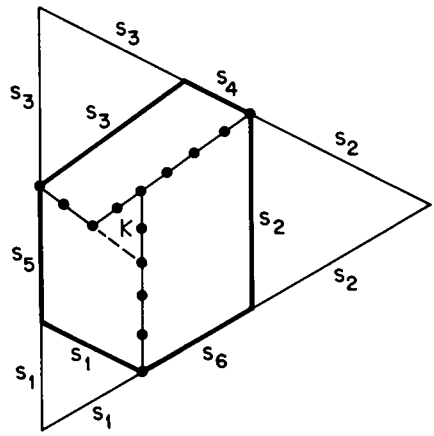


Fig. 8

ourselves to hex. For the upper bound on the bh, consider a *stretched* p-mino like Fig. 8, where $p = 14$, $K = 2$, $A = 9$, $B = 8$, and $C = 7$. We find

$$\begin{aligned} p - 1 &\geqslant \min(s_1, s_4) + \min(s_2, s_5) + \min(s_3, s_6) + 2|K| \\ &= \max(s_1, s_4) + \max(s_2, s_5) + \max(s_3, s_6) - |K|. \end{aligned}$$

Adding,

$$2(p-1) \geqslant \sum s_i + |K| = A + B + C + |K|,$$

or

$$\tfrac{1}{2}(A+B+C+|K|) \leqslant p-1. \tag{7}$$

For the lower bound on the bh, we have $p \leqslant \Delta$, the number of cells inside and on the boundary. To evaluate Δ, extend 3 sides of the bh into a triangle (see Fig. 8). Then

$$\Delta = \binom{s_1+s_5+s_3}{2} - \binom{s_1+1}{2} - \binom{s_2+1}{2} - \binom{s_3+1}{2}$$

Using (3), (4), and some manipulation, eventually

$$p \leqslant \Delta = (AB+BC+CA-1) - \tfrac{1}{4}[(A+B+C-1)^2 + K^2 - 1]. \tag{8}$$

Note that (7) and (8) are necessary and sufficient for a hex p-mino to exist with bh parameters A, B, C, and K, if they satisfy (6) already.

3. Symmetries

Let G_l be the symmetry group of the hex tessellation, that is, the group of all motions of the plane leaving the tessellation invariant, and let G_t be the normal subgroup of all translations in G_l. G_t is uninteresting because no finite configuration can be invariant under a translation, so we dispatch it in defining a fixed p-mino to be a class under G_t. A free p-mino is a class under all G_l. Under $G = G_l/G_t$ a fixed p-mino may transform into itself or into another fixed p-mino corresponding to the same free p-mino: the subgroup of G leaving it invariant is called its *symmetry*. Of course, conjugate subgroups of G describe the same symmetry in different orientations: so we define a *symmetry type* to be a conjugacy class of subgroups of G, and say that a free p-mino has such-and-such a symmetry type.

In the hex case G turns out to be D_6, the dihedral group of order 12. This is shown in Table I, with descriptions of the operations and suitable coordinate transformations. Referred to the origin, the latter are very simple: for example, a rotation through $\pi/3$ is simply $\mathbf{x} = (x, y, z) \to (-y, -z, -x) = \mathbf{f}(\mathbf{x})$. We need to refer them to the bh, and so have to add a translation $\mathbf{u}$ determined by requiring a specific corner $\mathbf{x}$ of the bh to transform into another specific corner $\mathbf{y}$, then solving for $\mathbf{u}$ in

$$\mathbf{y} = \mathbf{u} + \mathbf{f}(\mathbf{x}).$$

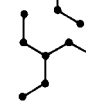
TABLE I

The Point Group G of the Hex Tessellation

Name	Effect	Transform $(x, y, z) \rightarrow$	Bh symmetry	Bh conditions
1	None	(x, y, z)	I	—
2	$\pi/3$ rotation	$(a+b'-y, b+c'-z, c+a'-x)$	G	$A=B=C, K=0$
3	$-\pi/3$ rotation	$(c+a'-z, a+b'-x, b+c'-y)$	G	$A=B=C, K=0$
4	$2\pi/3$ rotation	$(a-c+z, b-a+x, c-b+y)$	DD	$A=B=C$
5	$-2\pi/3$ rotation	$(a-b+y, b-c+z, c-a+x)$	DD	$A=B=C$
6	π rotation	$(a+a'-x, b+b'-y, c+c'-z)$	$R2$	$K=0$
7	$x=0$ reflection	$(a+a'-x, b+c'-z, b+c'-y)$	SD	$B=C, K=0$
8	$y=0$ reflection	$(c+a'-z, b+b'-y, c+a'-x)$	SD	$A=C, K=0$
9	$z=0$ reflection	$(a+b'-y, a+b'-x, c+c'-z)$	SD	$A=B, K=0$
10	x-axis reflection	$(x, b-c+z, c-b+y)$	D	$B=C$
11	y-axis reflection	$(a-c+z, y, c-a+x)$	D	$A=C$
12	z-axis reflector	$(a-b+y, b-a+x, z)$	D	$A=B$

TABLE II

The Symmetry Types of Hexagonal p-minos

Type	Index in G	Groups	Example
I	12	$\{1\}$	
$R2$	6	$\{1,6\}$	
S	6	$\{1,7\}\{1,8\}\{1,9\}$	
D	6	$\{1,10\}\{1,11\}\{1,12\}$	
$R3$	4	$\{1,4,5\}$	
SD	3	$\{1,6,7,10\}\{1,6,8,11\}\{1,6,9,12\}$	
R	2	$\{1,2,3,4,5,6\}$	
SS	2	$\{1,4,5,7,8,9\}$	
DD	2	$\{1,4,5,10,11,12\}$	
G	1	$\{1-12\}$	

Since we have referred a transformation to the bh, this must have sufficient symmetry to remain invariant, whether or not the enclosed p-mino does. So in the final columns of Table I we give the minimum bh symmetry necessary for the transformation to be performed at all, and the corresponding numerical criterion.

Table II lists the various types of symmetry possible for p-minos, which include the types for bh's. We give a name, the set of conjugate subgroups of G, and a hex example. The index of the subgroup in G equals the number of fixed p-minos corresponding to one free p-mino of that type.

By Statement 1 in Section 1 everything above applies to tri p-minos too, except that the examples in Table II are inapplicable where they occupy one ($p = 1$) or three colors. The apparently simple case of a single cell behaves rather oddly. Its symmetry type as a hex p-mino is G, but as a tri p-mino is only SS. The trouble is that a hex symmetry is only a tri symmetry as well if it maps the third color onto itself. This is only guaranteed provided $p \geqslant 2$, when both $+$ and $-$ must be occupied and so the third color can't map onto $+$ or $-$ under a symmetry.

4. Counting Algorithm

This is similar to that of [1]. Here we shall concentrate on the additional complications.

To start with, suppose we are to find $HX(p)$ by enumerating all fixed hex p-minos. Translations are eliminated by restricting the configurations to the triangular region $x \leqslant 0, y \leqslant 0, z \geqslant p$ (see Fig. 9) with $a = b = 0, 0 \geqslant (a, b) \geqslant -p$,

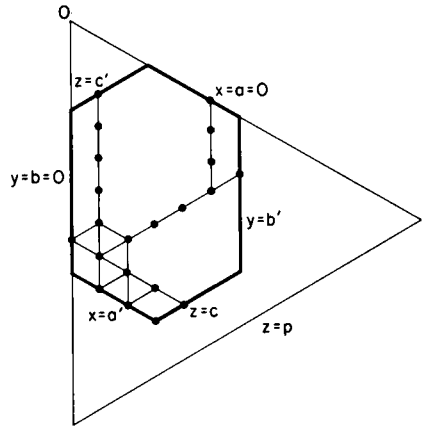


Fig. 9

and $0 \leqslant c' \leqslant c \leqslant p$. We insist on their touching the boundaries $x = 0$ and $y = 0$: that is, for the bh of a complete p-mino,

$$a = b = 0 \qquad (\text{when } q = p, \quad \text{see below}). \tag{9}$$

The choice algorithm is an elementary exercise in backtracking. At level $q-1$, $1 \leqslant q \leqslant p$, we have constructed some $(q-1)$-mino, and during level q we attach to it in turn all cells connected to it but unused at previous levels. From now on we shall take all the parameters $a, b, \ldots$ of the bh to apply to this partial configuration as well. Now rather than simply discard completed p-minos which fail to satisfy (9), we impose a further restriction on the choice of a qth cell, known as a *growth criterion*. In this simple instance it is

$$a = 0 \qquad \text{and} \qquad b \leqslant p - q; \tag{10}$$

that is, cell 1 is chosen on the x axis and subsequent cells are chosen close enough to the y axis to reach it by the time $q = p$.

In practice we only enumerate all free p-minos, computing $HX(p)$ from their symmetry types. First we fix the orientation of the bh, which may in general be in any of 12 positions, by insisting that (5) be satisfied. This is effected by the new growth criteria

$$(B \dot{-} A) - b \leqslant p - q \tag{11}$$

and

$$(B \dot{-} A) + (C \dot{-} B) \leqslant p - q \tag{12}$$

and

$$(\max(B, A) \dot{-} C) + (0 \dot{-} K) \leqslant p - q. \tag{13}$$

Here $x \dot{-} y = \max(0, x - y)$.

These are derived by noticing that, if a bh fails to satisfy (5) and (9) in more than one particular, the addition of a new cell can often only relieve one of its shortcomings at a time (see Fig. 9). If in (11) $b < 0$ and $B > A$, then attaching a new cell on the edge $y = b'$ to decrease b' will also increase B, so cannot decrease $B - A$. If in (12) $B > A$ and $C > B$, a new cell can increase $B - A$ or $C - B$ by 1 but not both. If in (13) B or $C > A$ and $K < 0$, a new cell can increase A by decreasing a', since $a = 0$, but cannot simultaneously increase s_1 (the length of side $x = a'$) or $K = s_1 - s_4$.

If the bh has no symmetry when $q = p$ (the usual case?) we are done. If not, the p-mino inside may have less symmetry than its bh, and we must define a canonical orientation inside its bh. Let the cells of the bh be enumerated in some fixed order, for example, in increasing order of x then y (see Fig. 10).

Let the *weight* of a fixed p-mino be a number, the binary expansion of

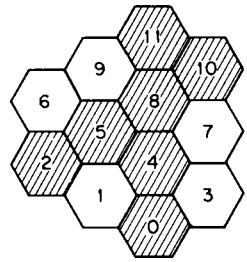


Fig. 10. A 7-mino (shaded) inside its bh.

which has digit i equal to 1 if cell i is a cell of the p-mino, and equal to 0 otherwise. Then a fixed p-mino is in canonical form, and counts as a free one, if it has weight at least as great as all its transforms by symmetries of the bh. In Fig. 10, bh symmetry is DD, 7-mino symmetry is D, weight of self is 101011001011, and weights of transforms are 100111101001 and 001111101010, respectively.

For tri p-minos, $p > 1$, the colors of the first two cells chosen are noted and future cells $q > 2$ chosen only from those two colors.

5. Performance, Results, and Omissions

An algorithm along the lines of Section 4 was written in ALGOL and run background on the Chilton Atlas I for 6 hr each on tri and hex, reaching $p = 16$ and 12, respectively. With machine code and 100-hr runs it should be possible to reach $p = 22$ and 16. Time is proportional to the size of the answer, and is exponential in p. The tri mode is slower per p-mino counted because of the increased depth of recursion with larger p.

Tables III–V show our results, with the square p-mino totals for completeness. Table III corrects Klarner [3] and Read [7], who both (!) found $HE(6) = 83$. $HX(p)$ has been confirmed for $p \leqslant 10$ by M. F. Sykes, by hand, in a single afternoon (!!). Table IV gives $TE(9)$ and $TE(10)$ larger by 1 and 4 than Read [7], because he is counting only simply connected objects.

Future computations ought to produce more detailed results, breaking the totals down by symmetry and bh as in [1], by number of free edges (of interest to theoretical physicists) or by connectivity (numbers of internal holes and boundary loops).

6. Asymptotic Behavior

In Tables III–V, we have included the ratios of successive fixed totals. The conjecture is irresistible that these approach limits, and the same for the free.

TABLE III

Hex p-mino Totals, Free and Fixed

p	$HE(p)$	$HX(p)$	Ratio	p	$HE(p)$	$HX(p)$	Ratio
1	1	1	1.0000	7	333	3652	4.4865
2	1	3	3.0000	8	1448	16,689	4.5698
3	3	11	3.6667	9	6572	77,359	4.6353
4	7	44	4.0000	10	30,490	362,671	4.6882
5	22	186	4.2273	11	143,552	1,716,033	4.7317
6	82	814	4.3763	12	683,101	8,182,213	4.7681

TABLE IV

Tri p-mino Totals, Free and Fixed

p	$TE(p)$	$TX(p)$	Ratio	p	$TE(p)$	$TX(p)$	Ratio
1	1	2	1.0000	9	160	1838	2.7230
2	1	3	1.5000	10	448	5053	2.7492
3	1	6	2.0000	11	1186	14,016	2.7738
4	3	14	2.3333	12	3334	39,169	2.7946
5	4	36	2.5714	13	9235	110,194	2.8133
6	12	94	2.6111	14	26,166	311,751	2.8291
7	24	250	2.6596	15	73,983	886,160	2.8425
8	66	675	2.7000	16	211,297	2,529,260	2.8542

TABLE V

Square p-mino Totals, Free and Fixed

p	$PE(p)$	$PX(p)$	Ratio	p	$PE(p)$	$PX(p)$	Ratio
1	1	1	1.0000	10	4655	36,446	3.6777
2	1	2	2.0000	11	17,073	135,268	3.7115
3	2	6	3.0000	12	63,600	505,861	3.7397
4	5	19	3.1667	13	238,591	1,903,890	3.7637
5	12	63	3.3158	14	901,971	7,204,874	3.7843
6	35	216	3.4286	15	3,426,576	27,394,666	3.8022
7	108	760	3.5185	16	13,079,255	104,592,937	3.8180
8	369	2725	3.5855	17	50,107,911	400,795,860	3.8320
9	1285	9910	3.6367	18	192,622,052	1,540,820,542	3.8444

Several authors, notably Klarner [3], have investigated this question theoretically. All that has been proved is that, by subadditivity, there exists

$$FL = \lim_{p\to\infty} (FX(p))^{1/p}, \tag{14}$$

where F stands for P, H, or T. If the fixed ratio limit exists, then the two are equal:

$$FL = \lim_{p\to\infty} FX(p)/FX(p-1), \qquad \text{if it exists.}$$

As to the free ratio, it is shown with some difficulty in [5] that, for square p-minos, the proportion of p-minos with any symmetry (other than I) approaches zero (but rather weakly), whence PE and PX behave the same in the limit but for a factor of 8. Following a general principle, the fixed totals are a little more tractable theoretically and numerically, which is why we choose to work with them.

The known bounds on the limits FL—or *critical points*, to borrow a sonorous phrase from theoretical physics—are

$$\begin{aligned} 4 < \mathrm{HL} &\leqslant 6.75, \\ 2.13 < \mathrm{TL} &\leqslant 4, \\ 3.72 < \mathrm{PL} &\leqslant 4.65. \end{aligned} \tag{15}$$

These are mainly due to Eden [4] Klarner [3, 10]; however, the upper bound on HL seems to have escaped previous notice, so here is the proof.

Let each cell have r edges, $r = 3, 4, 6$. We shall construct an encoding of fixed p-minos; suppose we are given a particular one. Starting from some fixed edge (the *root*) of some fixed cell (cell 1) of the p-mino, perform the following process on cell i for $i = 1, 2, \ldots$. Make up a string of binary digits $d_{ij} = 1$ if the jth edge from the root around the perimeter of cell i has an unnumbered cell of the p-mino adjoining, otherwise $d_{ij} = 0$. In the former case, give the cell the next unused number, and mark the common edge as its root. Omit the root edge of cell $i > 1$, since it must adjoin a numbered cell $j < i$. This procedure maps the p-minos one to one into the sequences with roughly p ones and $(r-2)p$ zeros. Hence

$$FX(p) \leqslant \binom{(r-1)p}{p} \simeq ((r-1)^{r-1}/(r-2)^{r-2})^p, \qquad \text{by Stirling's approximation,}$$

and

$$FL \leqslant (r-1)^{r-1}/(r-2)^{r-2}. \tag{16}$$

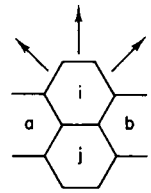


Fig. 11

For hexagons we can omit not only the root edge but also the two adjacent, since the cells a, b adjoining them also adjoin the root cell j, and must already have been investigated (see Fig. 11). So now the string has length $(r-3)p$, and instead of (16) we have

$$HL \leqslant (6-3)^{6-3}/(6-4)^{6-4} = 6.75. \tag{17}$$

Our own investigation will be empirical. We assume that the functions can be approximated by exponentials, and try to estimate the base γ and index β in the relation

$$FX(p) \sim \gamma^p p^\beta \times \text{constant}. \tag{18}$$

Then of course $\gamma = FL$. We hope shortly to describe our methods in a separate paper [9]. Our values are

$$\begin{aligned} HL &= \gamma_H \simeq 5.181 \pm 0.002, \\ \beta_H &\simeq -0.98 \pm 0.01, \\ TL &= \gamma_T \simeq 3.02 \pm 0.05, \\ \beta_T &\simeq -0.8 \pm 0.5, \\ PL &= \gamma_P \simeq 4.061 \pm 0.001, \\ \beta_P &\simeq -0.98 > 0.02. \end{aligned} \tag{19}$$

The triangular data behaves rather badly, and not just because of the small numbers involved. It is quite possible that actually $\beta = -1$ in all cases and there is an extra logarithmic factor. It is less likely that the γ are integers; γ_H is clearly not an integer.

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